

Skills Summary

Using Congruence to Solve for Corresponding Parts

Use this reference while completing your worksheet or when studying.

Every problem of this type is the same three moves: **(1)** read the congruence statement to find which parts match, **(2)** set those two measures equal (CPCTC), **(3)** solve the equation — and substitute back if the question asks for the measure rather than the variable.

Skill 1 — Corresponding sides

Correspondence: in $\triangle ABC \cong \triangle DEF$ the letters match by position: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. A side is named by two letters, so $\overline{AB} \leftrightarrow \overline{DE}$, $\overline{BC} \leftrightarrow \overline{EF}$, $\overline{AC} \leftrightarrow \overline{DF}$.

CPCTC: corresponding parts of congruent triangles are congruent, so $AB = DE$ — an equation you can solve.

Example: $\triangle ABC \cong \triangle DEF$ with $AB = 3x - 1$ and $DE = 14$. Find x .

Step 1. $\overline{AB}$ and $\overline{DE}$ are named by matching positions, so CPCTC makes their measures equal — write that equation.

Step 2. $3x - 1 = 14$, so $3x = 15$ and $x = 5$.

Try it: $\triangle GHI \cong \triangle JKL$ with $GH = 4x + 3$ and $JK = 27$. Find x .

check: $9 = x$

Skill 2 — Corresponding angles

Same rule, single letters: in $\triangle PQR \cong \triangle STU$, $\angle P \leftrightarrow \angle S$, $\angle Q \leftrightarrow \angle T$, $\angle R \leftrightarrow \angle U$, and corresponding angles have equal measures.

When one measure is missing: the three angles of either triangle add to 180 degrees, so find the missing angle in the triangle you know, then carry it across by CPCTC.

Example: $\triangle PQR \cong \triangle STU$ with $m\angle Q = 6x - 8$ and $m\angle T = 40$ degrees. Find x .

Step 1. Q and T sit in the same position in the statement, so their measures are equal by CPCTC.

Step 2. $6x - 8 = 40$, so $6x = 48$ and $x = 8$.

Try it: $\triangle PQR \cong \triangle STU$ with $m\angle R = 3y + 5$ and $m\angle U = 35$ degrees. Find y .

check: $10 = y$

Skill 3 — Reading the correspondence correctly

The order of the letters is the whole message. $\triangle ABC \cong \triangle DFE$ is *not* the same statement as $\triangle ABC \cong \triangle DEF$.

- Write the two names one above the other and pair off the columns.
- Pair the letters in the order given — never re-alphabetise the second triangle.
- Then a side pairs with the side named by the matching two columns.

Example: $\triangle ABC \cong \triangle DFE$ with $AB = 2x + 1$ and $DF = 19$. Find x .

Step 1. Pair the columns: $A \leftrightarrow D$, $B \leftrightarrow F$, $C \leftrightarrow E$, so $\overline{AB}$ matches $\overline{DF}$ (not $\overline{DE}$).

Step 2. $2x + 1 = 19$, so $2x = 18$ and $x = 9$.

Try it: $\triangle MNP \cong \triangle RTS$ with $MN = 5x - 2$ and $RT = 23$. Find x .

$x = 5$ ✓

(!) **Watch out:** solving for x is usually not the end. If the question asks for a *measure*, substitute your x back into the expression — and check it against the other expression, which must give the same value.